Generalized Linear Models

Introduction and Overview

Prof. Dr. Andreas Groll & Dr. Daniel Horn

Summer term 2026

Organizational stuff

Suitable modules: BS 14/15; MS 6/7; MD E1 (Methods); ME 7

Lecture

- inverted classroom with online videos
- Q&A session on Wednesday, 12-13 pm,
  lecture room: M/E 21

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Exercise

- Thursday, 12-14 pm, room M/E 25
- 1st exercise on 30.04.2026

Dr. Daniel Horn (dhorn@statistik.tu-dortmund.de)

Organizational stuff

Website

`https://moodle.tu-dortmund.de/course/view.php?id=57017`

Enrollment key: GLM-SoSe-26

Quiz

- 1st quiz on Wed, June 3, 2nd quiz on Wed, July 1, each 12:00 instead of Q&A sessions (there'll be different Q&A formats in theses weeks, since AG is not in Dortmund).
- registration via Moodle until 2 weeks before the respective quiz takes place

Exams

- 1st date: 10.08.26, 13:30-15:30, room EF 50 / HS1
- 2nd date: 09.10.26, 14:00-16:00, room HGII / HS6

Goals of this course

Get familiar with advanced regression techniques

- Generalized regression models to analyse panel data
- Categorical regression
- Variable selection via penalization
- How to deal with longitudinal data

Learning objective: acquire competences ...

... to choose, compare and interpret suitable models for a given data situation.

Content

- Based on the course from Annika Hoyer (and colleagues, LMU Munich)
- No complete manuscript, but:

$\Longrightarrow$ video lecture: a mixture out of slides & hand-written parts; Q&A sessions

Literature

- Fahrmeir, Kneib, Lang, Marx (2022): Regression: Models, methods and applications. Berlin. Springer-Verlag (3rd edition)
- Fahrmeir, Tutz (2013): Multivariate statistical modelling based on generalized linear models. Springer Science & Business Media
- Tutz (2011): Regression for categorical data. Cambridge University Press
- Eilers, Marx (2021). Practical smoothing: The joys of P-splines. Cambridge University Press.

Course overview

1. Basic concepts of regression

Course overview

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2. Binary regression

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models
4. Multi-categorical regression

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models
4. Multi-categorical regression
5. Semi- and non-parametric regression

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models
4. Multi-categorical regression
5. Semi- and non-parametric regression
6. Multivariate/ extended regression models

Survey

Please participate in this survey, it will help us to be able to better guide you in the future!

`https://umfragen.tu-dortmund.de/index.php/966481?lang=en`

